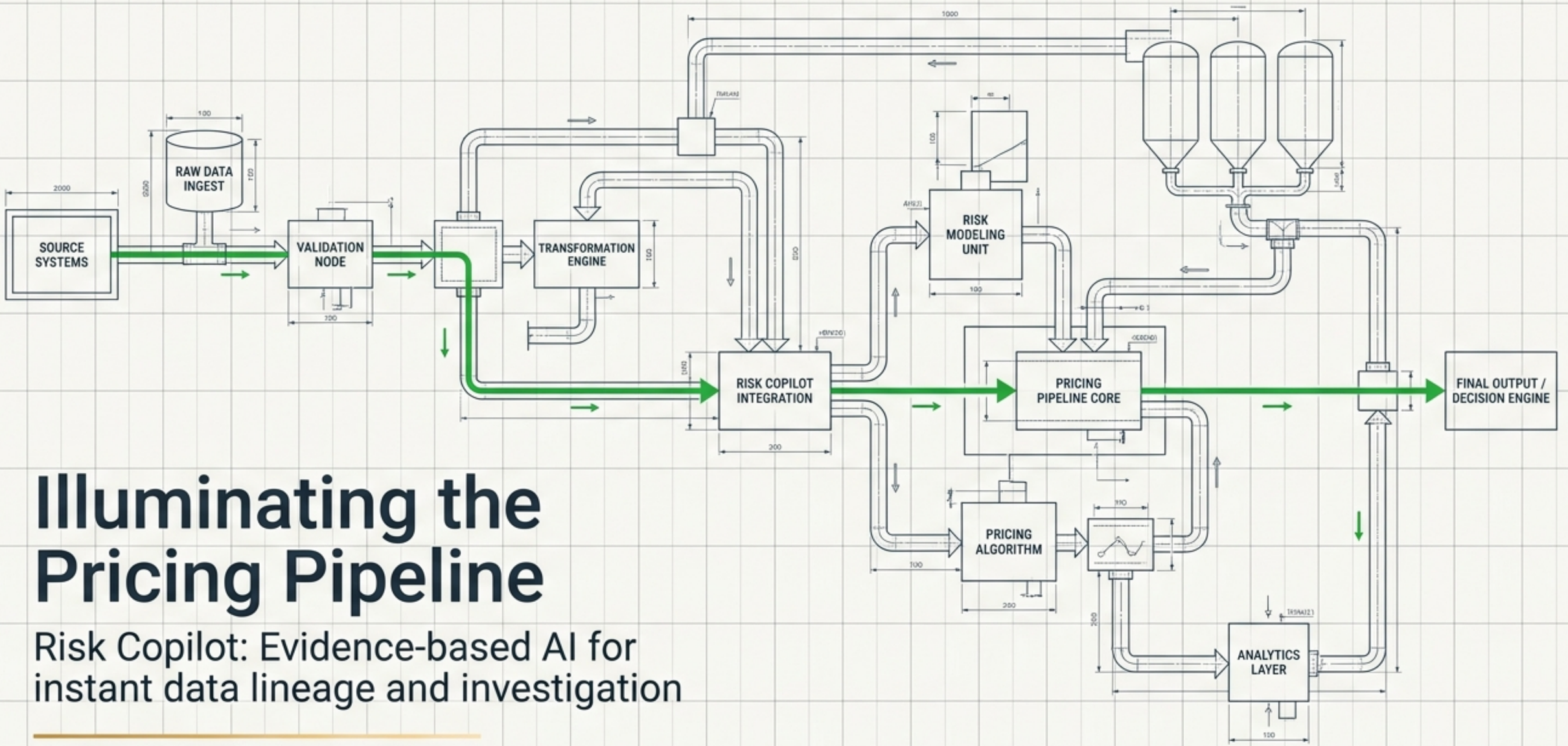
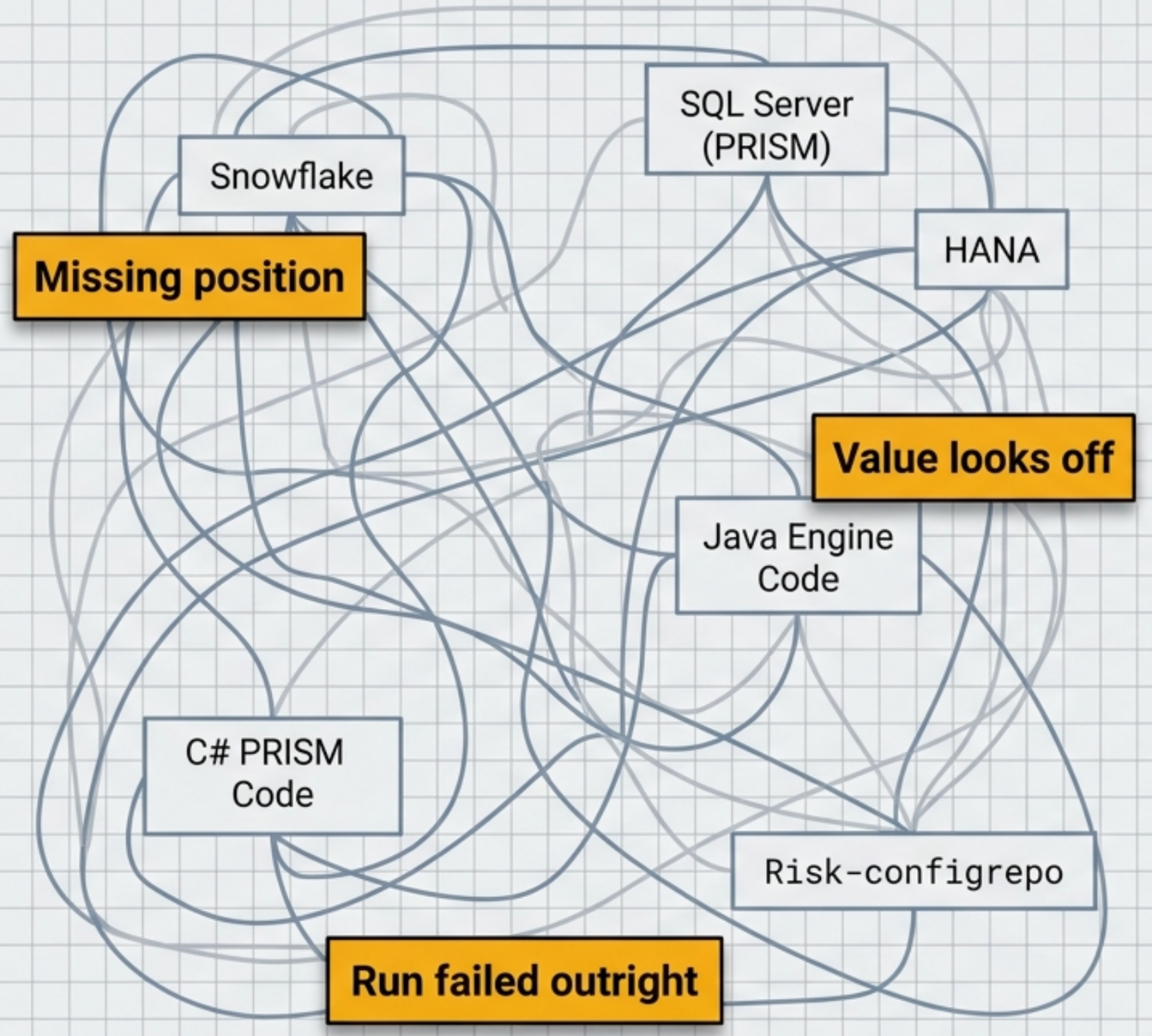


Illuminating the Pricing Pipeline

Risk Copilot: Evidence-based AI for instant data lineage and investigation

For Business Analysts & Developers

NotebookLM

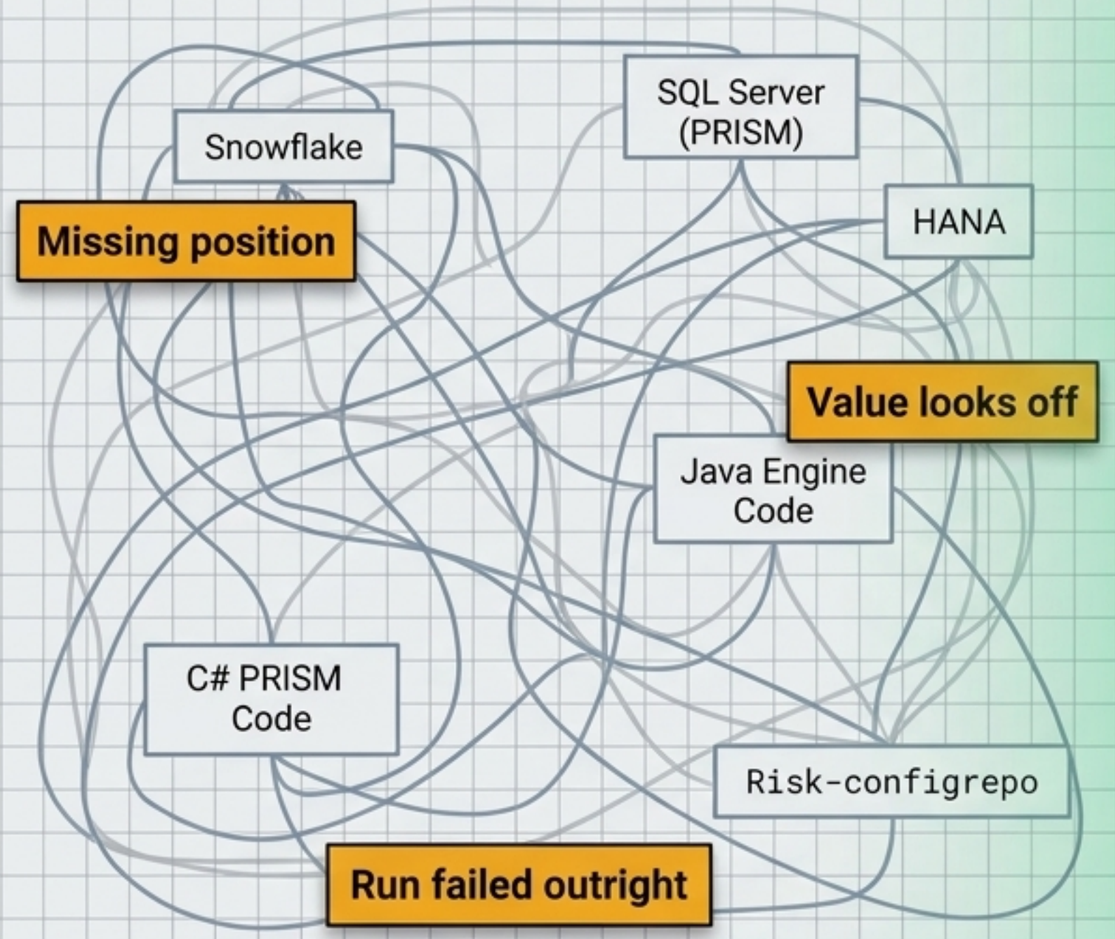


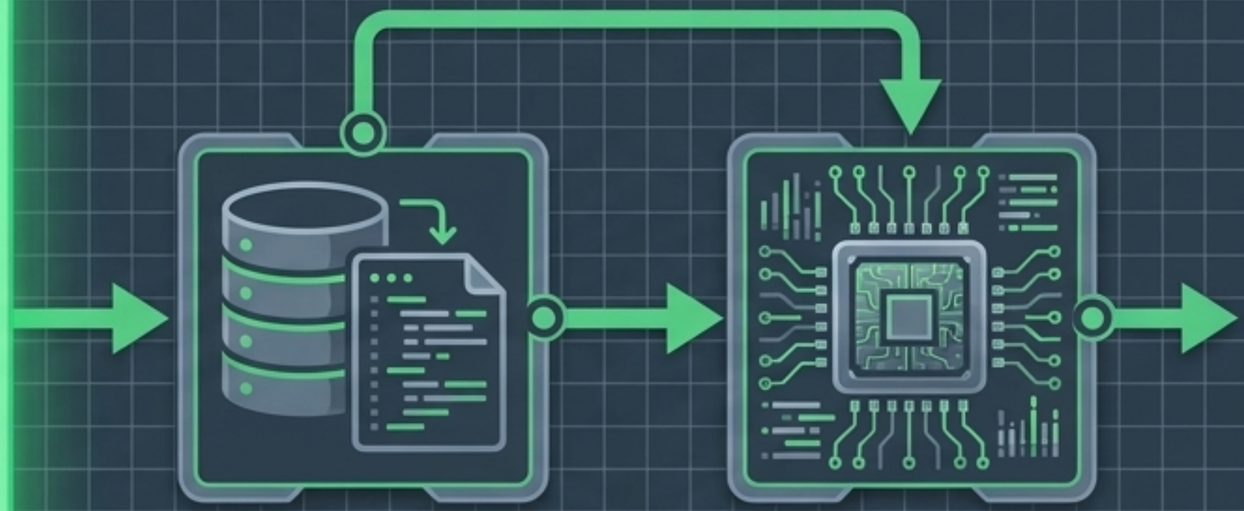
Investigating pricing failures today requires navigating a dark maze of disconnected systems and tribal knowledge.

- Who knows which engine, which staging table, which **fetch view**, and which property drives each hop?
- It is **slow** and **error-prone**.
- Relies on **knowledge locked in a few people's heads**.

NotebookLM

Risk Copilot acts as an X-ray flashlight, using Retrieval-Augmented Generation (RAG) to instantly trace data lineage.





Step 1: Retrieve

Pulls precise facts from private, indexed codebase (Java, PRISM) and live databases.

Step 2: Generate

The LLM processes retrieved facts to answer the user's natural language question.

Standard LLMs don't know our private codebase. Copilot solves this by retrieving exact code, properties, or query results first, ensuring answers are grounded in real evidence, not guesses.

NotebookLM

Meet the Specialists: Pre-briefed agents equipped with custom playbooks for our specific landscape



@vsc-troubleshooter

Target Domain:
VSC Engines (RISK.VSC.ENGINES)

Data Traced:
Source DB → VSC.STG_* → VSC.V_STG_* → Engine → Results

Core Capability:
Reads UTILS.APPLICATION_LOG and VSC.LOG_POSITIONDATA

Prompt Example:
"Why is security_ik 12345 missing from the fixed rate bond results for run 45678? "



@prism-troubleshooter

Target Domain:
PRISM Engines (LST.MAIN.CLOUD)

Data Traced:
HANA/Snowflake → STG_GS_* → Engine → Result_<Engine>

Core Capability:
Checks Scenario_Execution and Execution_Log for failures

Prompt Example:
"The FX Barrier Option staging failed for 2026-06-18 – what happened?"



@engine-property-resolver

Target Domain:
The Property Layer

Data Traced:
Crosses -default, -uat, and -dev config files

Core Capability:
Extracts target SQL and runs read-only queries with correct parameters

Prompt Example:
"How does the cross-ccy swap engine fetch its positions?"



@git-repo-helper

Target Domain:
Azure DevOps Repos

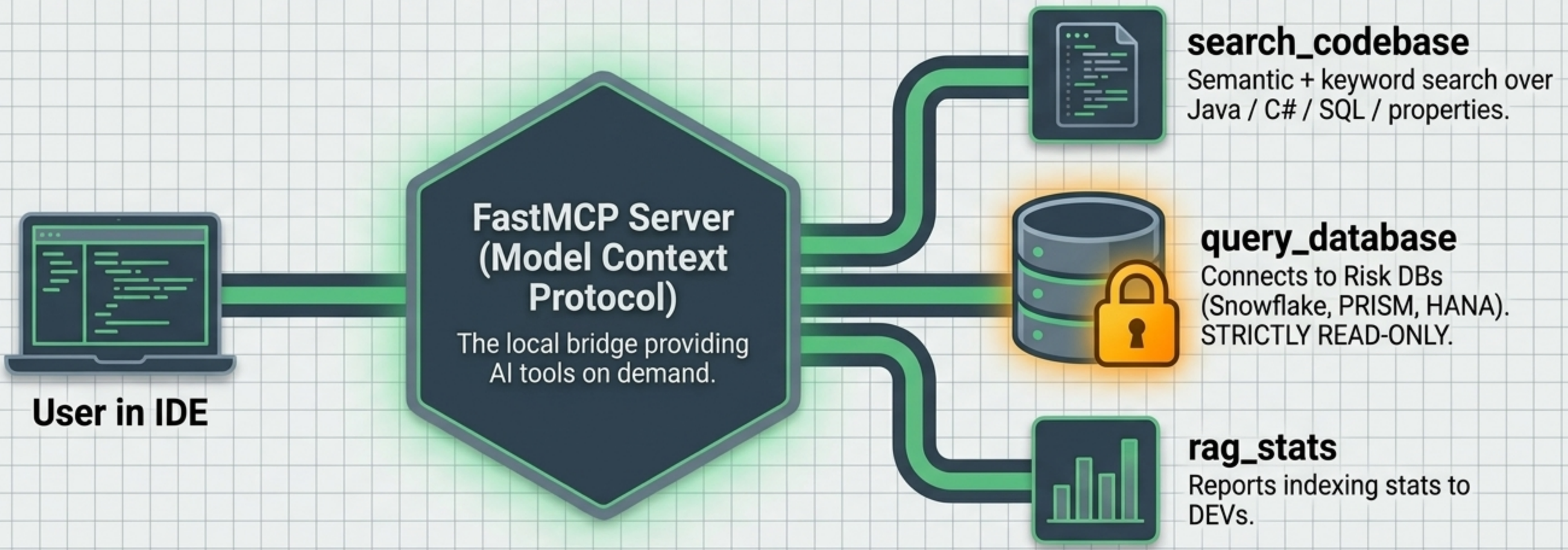
Data Traced:
Branch history and text search

Core Capability:
Clone, pull, diff, and branch switching via chat

Prompt Example:
"Search the repo for..."

NotebookLM

The AI Bridge: How GitHub Copilot securely accesses our private infrastructure.

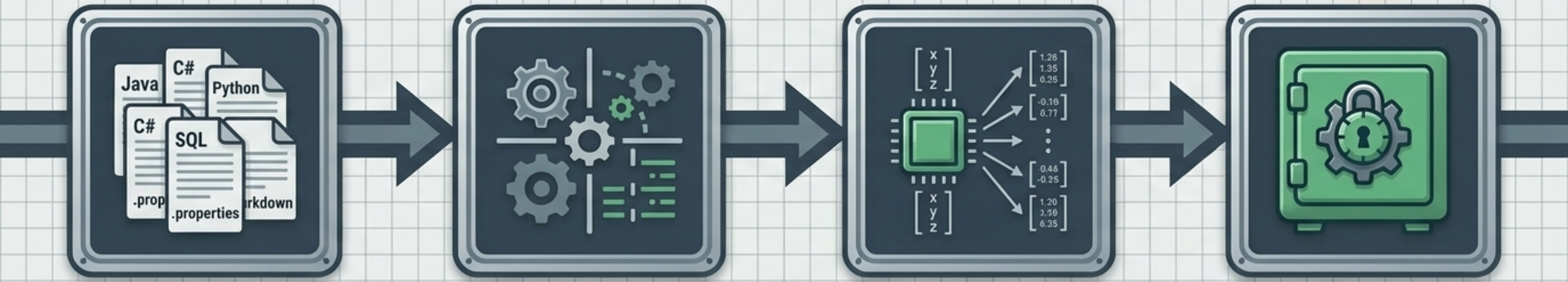


Incremental Freshness:
Tracked repos are git-pulled and re-indexed incrementally so only changed files are processed.

NotebookLM

The Engine Room: Translating 56,800+ files into searchable meaning, fully locally.

100% Local Execution.
The model, vectors, and index never leave your machine.



Raw Code Intake

Java, C#, Python, SQL, .properties, Markdown.

Tree-sitter (Parsing)

Parses into logical structures (whole methods, whole views). Understands true structure, not arbitrary text slices.

all-MiniLM-L6-v2 via sentence-transformers

A fast, ~80 MB CPU model that converts text chunks into arrays of numbers (vectors) capturing intent.

ChromaDB

Persistent local vector database holding ~56,800 indexed chunks.

NotebookLM

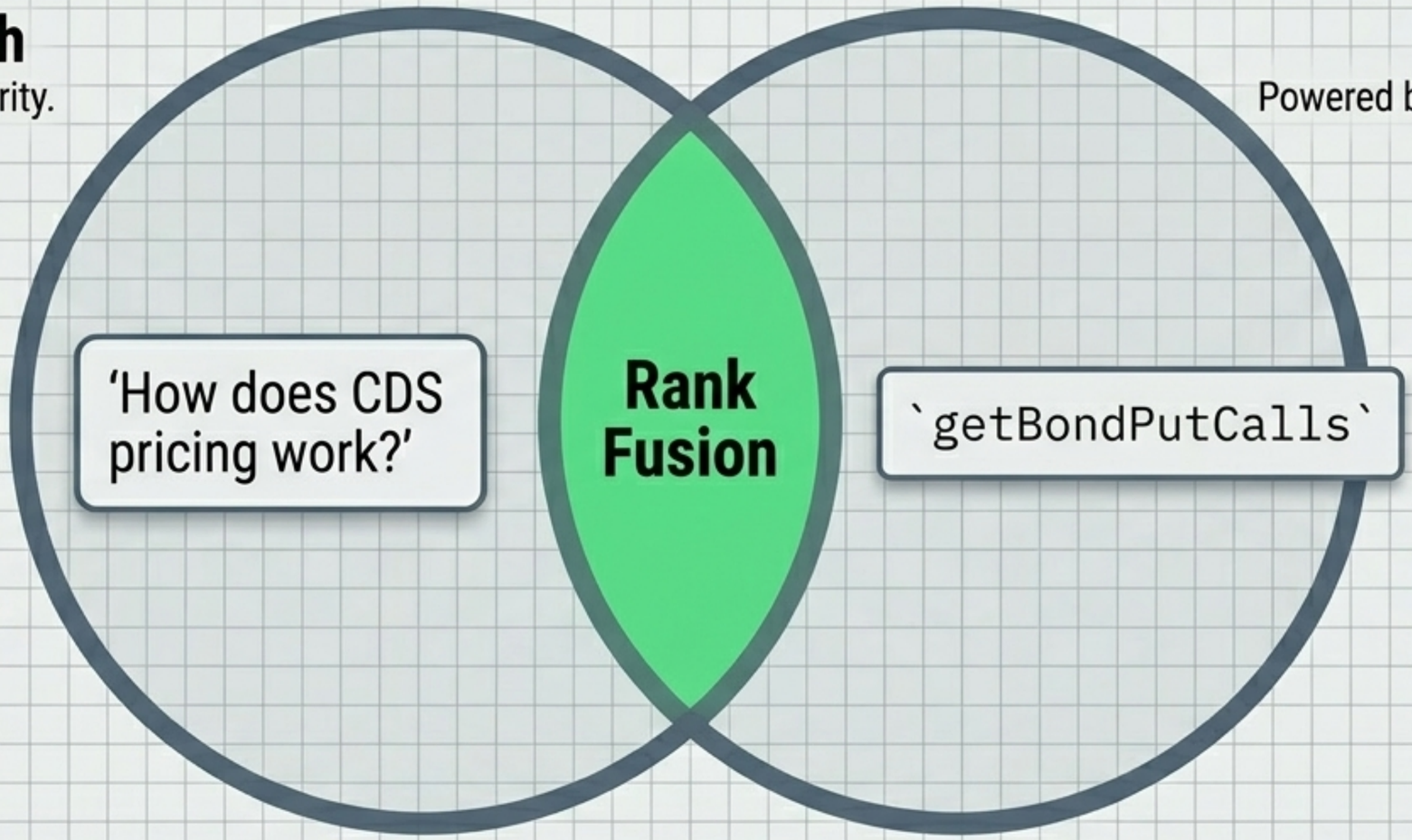
The Search Net: Merging conceptual intent with precise technical terminology.

Semantic Search

Powered by Vector Similarity.
Captures Meaning.

BM25 Search

Powered by exact term matching.
Captures Keywords.



Meaning-based search excels at conceptual exploration, while keyword search locks onto exact symbol names. Together, they create an inescapable contextual net for the AI.

NotebookLM

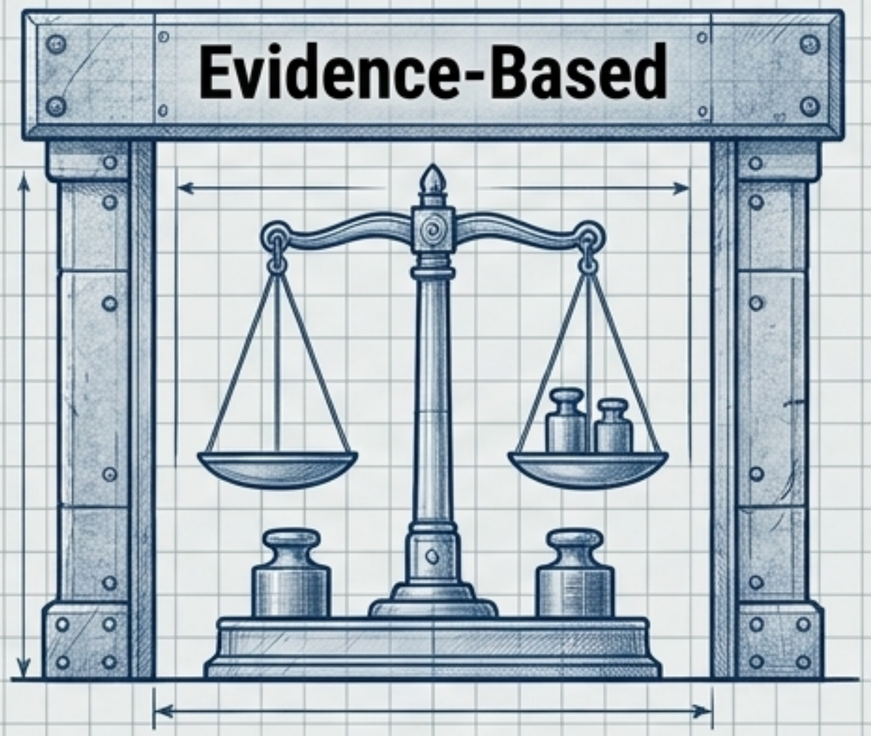
Enterprise AI demands absolute certainty: Our three pillars of Trust & Truth.



Safe by Design

100% Local execution.

Database access via query_database permits SELECT operations only. UPDATE, DELETE, and INSERT are structurally impossible.

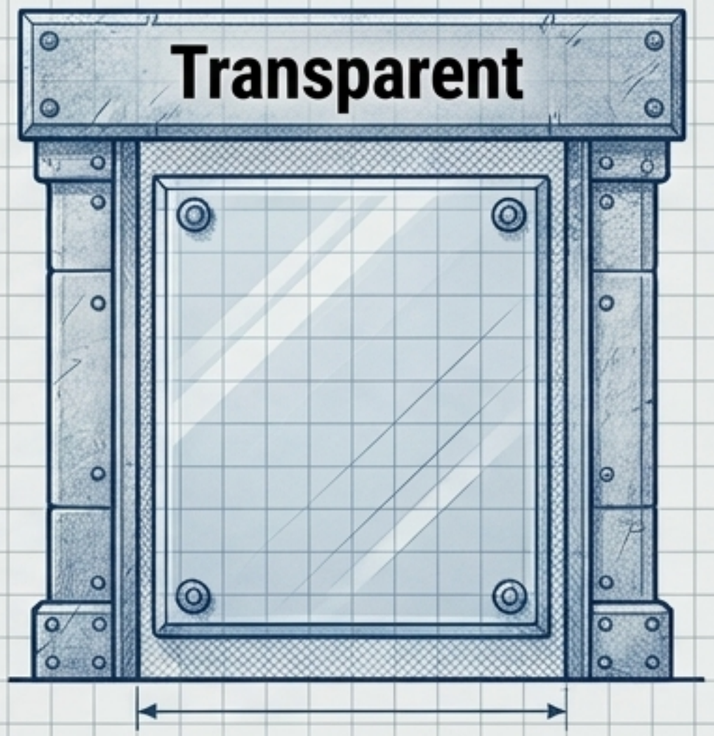


Evidence-Based

Rules of engagement: Outputs are explicitly tagged.

- **Confirmed:** Found actual source code or query result.
- ⚠️ **Inferred:** Deduced from indirect evidence.

If data is missing, it suggests next steps rather than inventing gaps.



Transparent

No black boxes.

Before any database results appear, the exact SQL query is always echoed back to the IDE so you can verify precisely what was run.

NotebookLM

Scaling our investigations through convention, not hard-coding.

- **Velocity:** Minutes instead of hours to answer “where did this position go?”
- **Democratization:** Lowers the knowledge barrier so BAs can self-serve data-lineage.
- **Reliability:** Evidence-based certainty completely replaces guesswork.
- **Scalability:** New engines are discovered via search conventions, requiring zero manual updates.

> **Risk Copilot turns our own code and databases into something you can simply ask, delivering trustworthy, evidence-backed answers in seconds.**

NotebookLM

Ready to investigate? Paste these exact prompts into Copilot today.

For Everyone / BAs

```
@vsc-troubleshooter Why is security_ik
12345 missing from the fixed rate bond
results for run 45678?

@engine-property-resolver How does the
cross-ccy swap engine fetch its
positions? Run it for 2024-12-31.

What's the latest effective date in
the bond coupon staging view?
```

More Technical / DEVs

```
@prism-troubleshooter The FX Barrier
Option staging failed for 2026-06-18 —
— what happened?

Find the getBondPutCalls method.

How much code is currently indexed?
```

See `.github/agents/*.md for full playbook setups.

NotebookLM